



The Air Fryer Market

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Air Fryer Market Analysis

Data check and market trend findings from the cleaned brand-year dataset

50 rows

balanced brand-year observations

10 brands

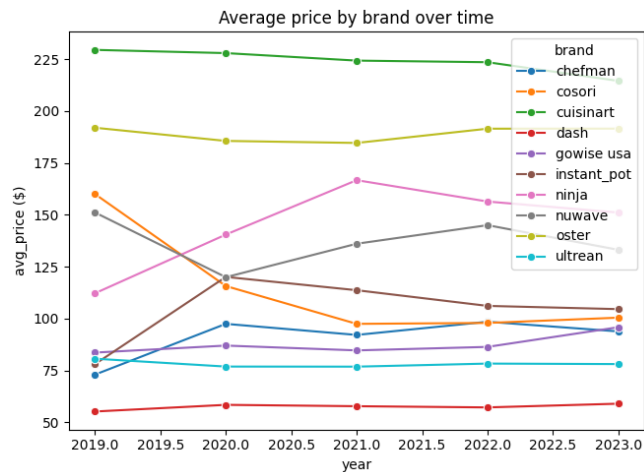
major air fryer brands

2019-2023

years included in the sample

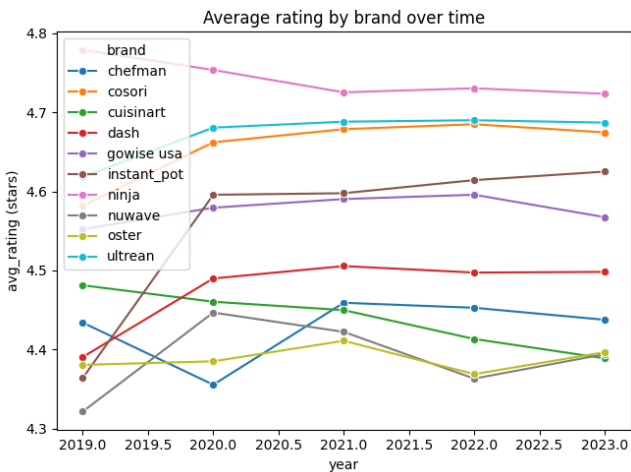
1.0 total share

shares add up within each year



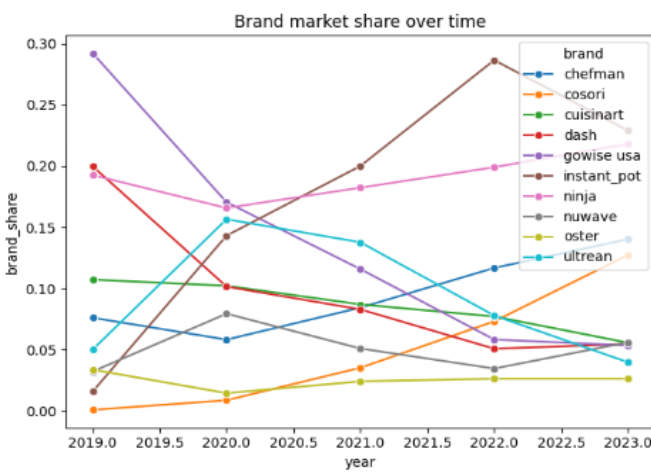
Price

Brands separate into clear price tiers, with premium brands sitting well above value-oriented competitors.



Rating

Ratings are tightly clustered around high values, so ratings alone do not explain large market-share differences.

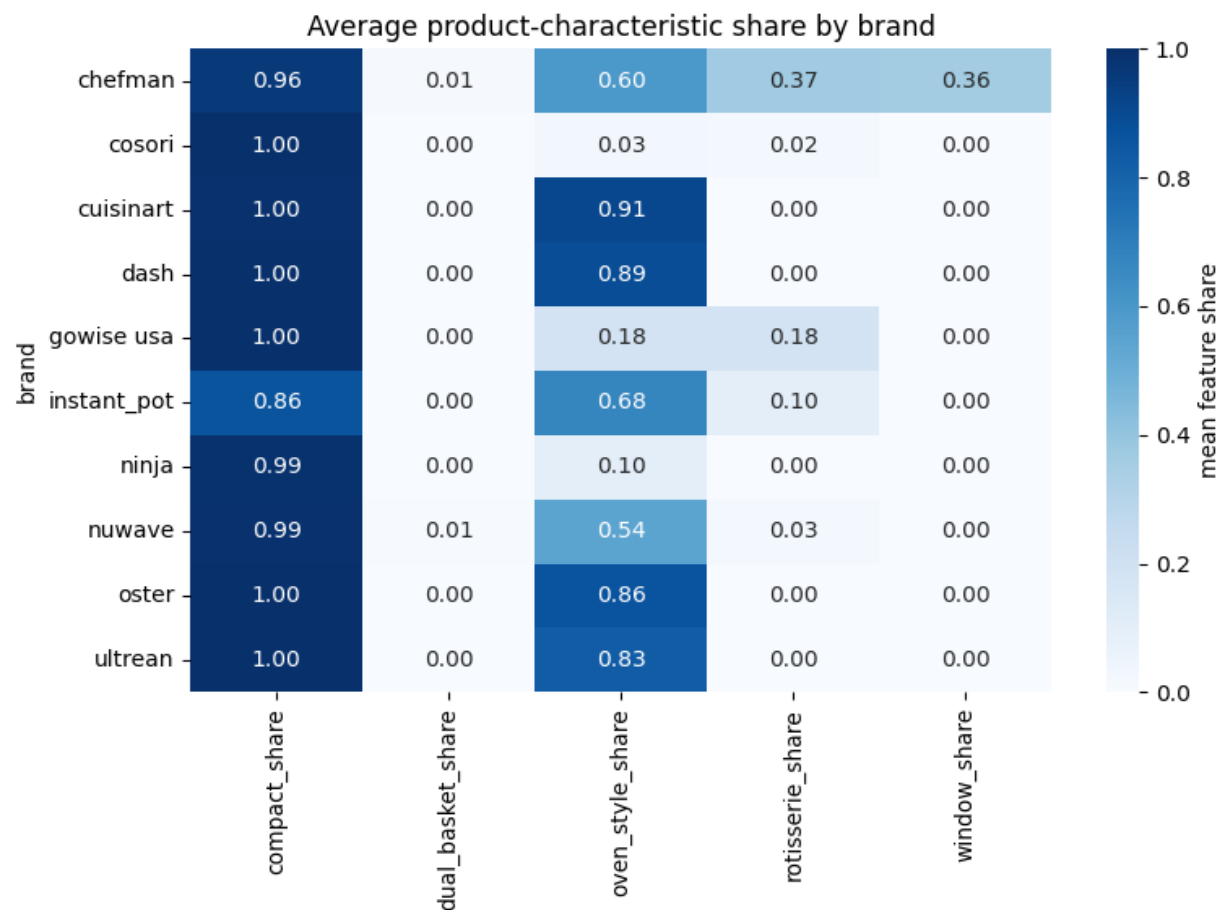


Market Share

Market shares seem to start off quite scattered with a select few growing while several fall below 5% with the rest sitting somewhere in between.

Product characteristics and main takeaway

Market is segmented, with concentrated leaders, visible price tiers, and feature-based differentiation



Feature pattern

Compact designs appear across almost the entire market, while oven-style, rotisserie, window, and dual-basket features differ more by brand.

Brand positioning

The heatmap here implies that the brands are not just competing on price, but on product format and feature bundles to target the different segments of customers.

Connection to next section

The descriptive results set up the demand model, where log brand share becomes the outcome and price, rating, features, brand effects, and year effects become predictors.

Demand Model: Estimating How Price and Features Affect Market Share

Model setup and key variables used to explain brand market share

Outcome Brand-year market share	Model Multinomial logit-style specification	Predictors Price, rating, and features	Fixed Effects Brand and year controls
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Model Overview We estimate demand using a multinomial logit-style model where the outcome is the log of each brand’s market share. The goal is to see how price, ratings, product features, brand identity, and year trends are related to market share.	Equation $\log(\text{share_brand_year}) = \beta_0 + \beta_1 \text{ price} + \beta_2 \text{ rating} + \beta_3 \text{ features} + \text{brand effects} + \text{year effects} + \varepsilon$ Predictors • Average price • Average rating • Product features • Brand fixed effects • Year fixed effects
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Overview The model estimates how multiple brand, product, and time factors relate to market share using a multinomial logit framework.	Specification Market share in logs is modeled as a function of price, rating, product features, and brand and year fixed effects.	Interpretation Coefficients indicate how each factor is associated with higher or lower market share after controlling for brand and year differences.
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This model helps explain which factors are associated with higher or lower brand share after controlling for brand and year differences

Demand Model: Estimating How Price and Features Affect Market Share

Model setup and key variables used to explain brand market share

Price Negative, significant effect	Rating Positive, significant effect	Features Mostly not significant	Controls Brand and year effects included
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Model Overview Demand is estimated using log market share. The model controls for price, ratings, product features, brand effects, and year effects.	Key Results Price: $\beta = -0.028$ Statistically significant Rating: $\beta = +9.17$ Statistically significant	Interpretation Higher prices are associated with lower demand. Higher-rated products are associated with higher demand. Product features have limited impact.
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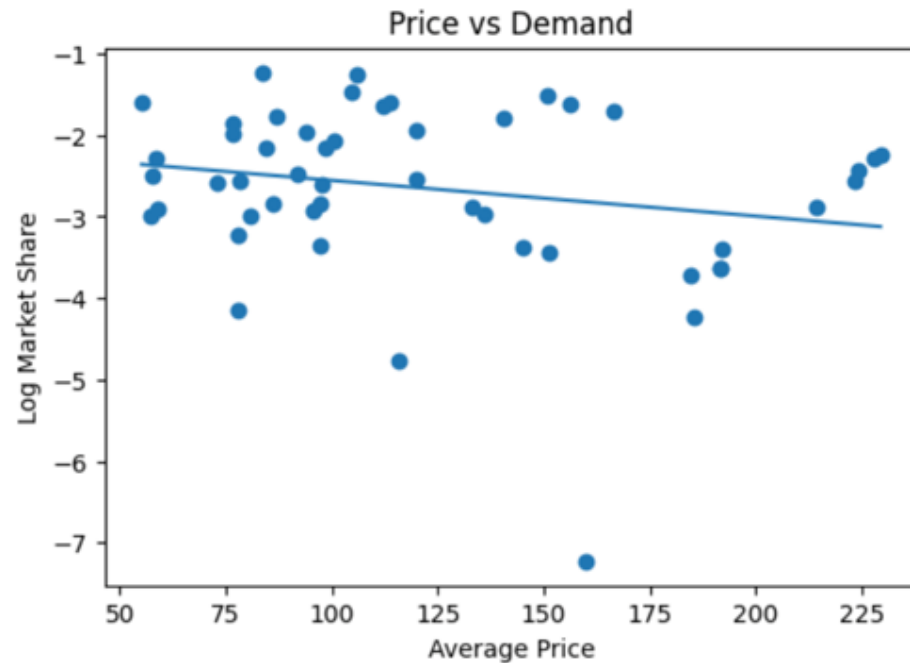
Price The negative price coefficient suggests that demand falls when average price rises, holding other factors constant.	Ratings The positive coefficient suggests that perceived product quality is strongly related to demand.	Product Features Most feature variables are not statistically significant, suggesting features explain less variation than price and ratings.
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Demand is primarily driven by price and ratings, while product features have limited impact on demand

Price vs Demand

Evidence that higher prices are associated with lower market share

Scatterplot: Average Price and Log Market Share



Key Findings

- Clear negative relationship between price and market share
- Consistent with regression result ($\beta_{\text{price}} < 0$)
- Consumers appear to be price sensitive

Interpretation

As price increases, demand decreases. This is consistent with economic theory and supports the model result.

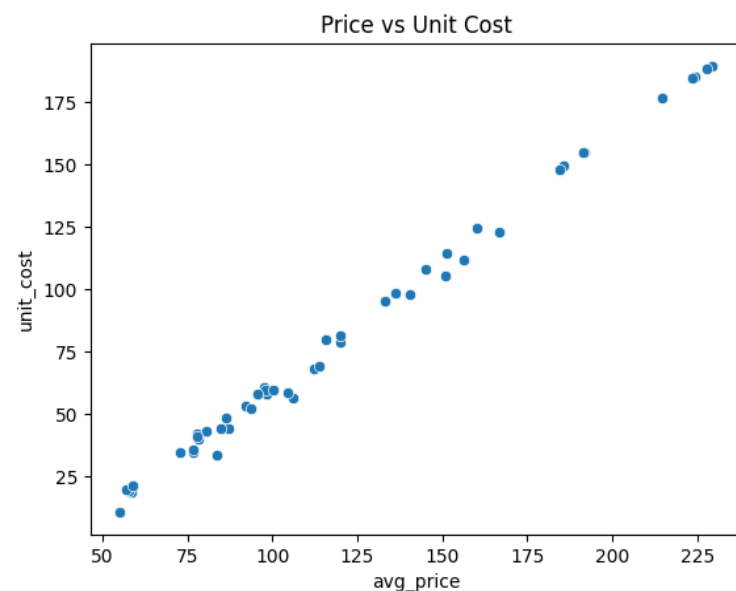
Price is a meaningful driver of demand because higher prices are linked to lower brand share.

Pricing Analyst Results

Costs, markups, and profitability patterns by brand

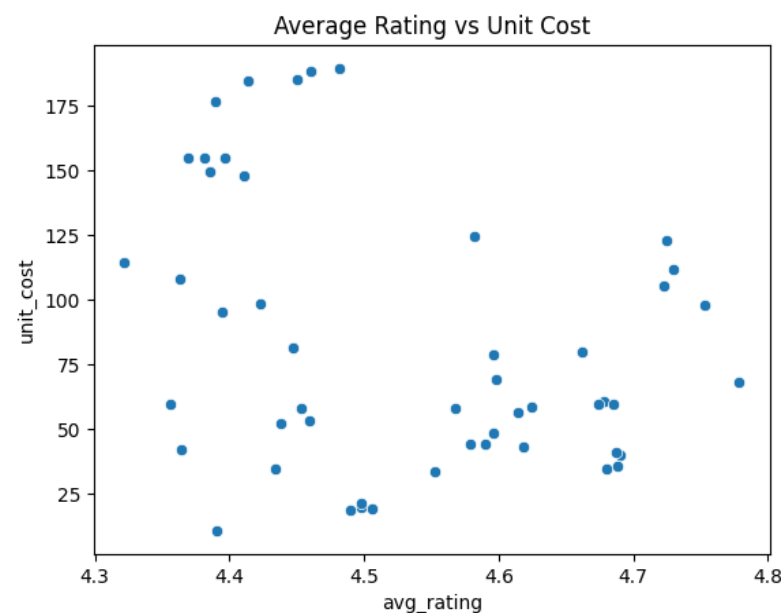
Price and Cost

Higher prices imply higher unit costs



Ratings and Cost

Ratings do not clearly increase with cost



Highest Cost Brands

Cuisinart, Oster, and Ninja

Meaning

Higher-priced brands appear to have higher inferred costs, but higher cost does not automatically mean higher customer satisfaction.

Pricing position matters, but strong ratings and profitability require more than simply charging a higher price.

Market Positioning and Conclusion

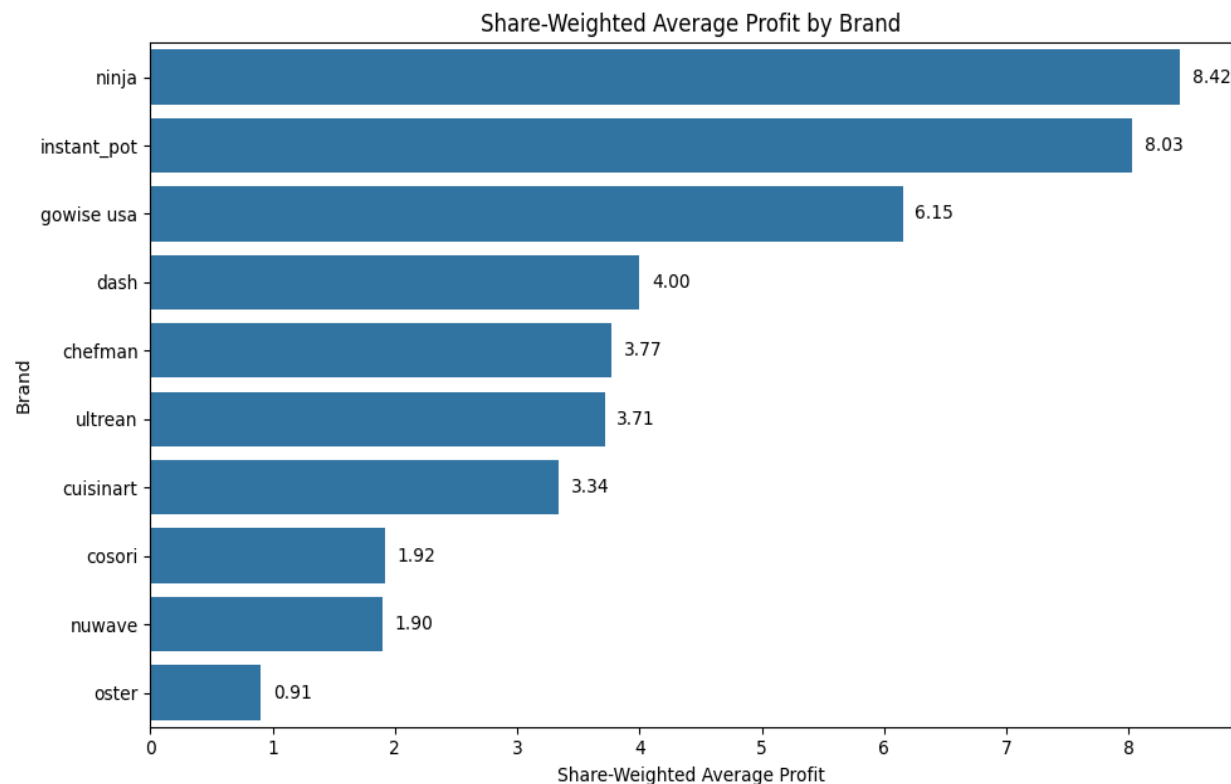
Brand positioning and business takeaways

Main Takeaways

- Ninja has the highest share-weighted average profit: 8.42
- Instant Pot (2nd): 8.03
- GoWISE USA (3rd): 6.15
- Oster (lowest): 0.91

Conclusion

Ninja and Instant Pot look strongest because they combine high markups with stronger market share. Cuisinart and Oster are premium or high-cost brands, but they are not the most profitable by share-weighted profit.



Bottom line: Profitability is driven by both markup and market share, not price alone.